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Spatio-temporal distribution of cigarette butt contamination in urban beaches with varying levels of use

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ABSTRACT

This study assessed the extent of cigarette butt contamination on two urban beaches in Recife-PE (Pernambuco State, Brazil) with varying levels of use. The study considered degradation levels and evaluated whether the brands varied by time, space, and beaches use. Ten transects (1.5 m wide) and spaced 10 m apart, were delimited on the beaches investigated. Twelve collections were made between May 2021 and January 2022, and the collected cigarette butts were evaluated based on degradation level, mass, size, and brand. In total, 10,275 cigarette butts were collected on both beaches, with P1 accounting for 96.91% of the found cigarette butts. The density of cigarette butts on the beaches was directly proportional to the level of use, with 8.85 butts/m^2 in P1 and 1.05 butts/m^2 in P2. Eighteen brands were identified, with brand (**A**) being the most popular regardless of the area. Significant differences ($p \leq 0.05$) were found when evaluating the amount of butts/m^2; Sundays with high precipitation presented reduced values of butts; more occupied areas presented transects with higher amounts; abundance in the summer; morphometry with higher values recorded in newly discarded butts; predominance of more degraded butts and brand diversity. Despite the difference in the number of butts/m^2 between the areas, it was possible to identify that their abundance is very expressive, and that the monitored beaches present a high exposure to the contaminant.

1. Introduction

Cigarette butts, linked to the tobacco industry or smokers, contain over 150 toxic compounds among the more than 7,000 present in their structure. A single cigarette butt can contaminate approximately one thousand litres of water (Torkashvand et al., 2021; Araújo and Costa, 2019). Furthermore, it is estimated that tobacco production worldwide requires 22 billion tons of water, a process that generates over 80 million tons of carbon dioxide annually (United Nations, 2022). Research analysing the impact of cigarette butts on water bodies has identified compounds such as nicotine, cadmium, copper, lead, chromium, arsenic, and polycyclic aromatic hydrocarbons (Green et al., 2014; Valcarcel et Al., 2011; Buerge et al., 2008; Moriwaki et al., 2009).

Its impact on biodiversity can occur directly, through the ingestion of cigarette butts by marine organisms, causing their mortality by starvation (Macedo et al., 2011); or by indirect contact, from the release of compounds through leaching resulting in decreased mobility, modification of heart rate, modification of DNA structure, and mortality in biota (Mansouri et al., 2020; Rawls et al., 2011; LEE and LEE, 2015; Booth et al., 2015). In addition, compounds such as pesticides, herbicides, and insecticides can also have chronic and acute effects through the introduction of metals into tobacco leaves (Zoffoli et al., 2013).

Reports from cleaning campaigns indicate that the improper disposal of cigarette butts is becoming increasingly prevalent, with these items being the most commonly collected waste. In 2020, over 900,000 cigarette butts were collected from coastal regions worldwide (Ocean Conservancy, 2022). Of the 6 trillion cigarettes consumed annually, 4.5 trillion are discarded improperly into the environment (Novotny and Slaughter, 2014). This issue is widespread across the globe, with cigarette butts accounting for up to 85% of the total items collected on beaches in Germany, 59.4% in Cyprus, and between 38–41% in marine areas of Chile. Additionally, on the Mediterranean coast of Israel, cigarette butts made up between 1.4–30.6% of the total items collected along eight beaches (Kataržytė et al., 2020; Loizidou et al., 2018;

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Hidalgo-Ruz et al., 2018; Pasternak et al., 2017).

In the coastal area of Recife, Brazil (approximately 8 km), cigarette butts can account for up to 14.95% of marine litter (Silva-Cavalcanti et al., 2013). Studies conducted in this region have revealed that the high number of improperly disposed items is largely influenced by the large influx of tourists, high occupancy rates, and inadequate cleaning services (Araújo et al., 2022; Silva-Cavalcanti et al., 2013; Dias-Filho et al., 2011). Despite the removal of 20 tons of trash daily from the beaches in this area (EMLURB, 2022), this figure does not include the cleanup of cigarette butts, because the equipment used is not capable of collecting items between 2.1 and 2.7 cm in length, as the machinery and manual cleaning service cannot pick up this size.

The issue of marine litter is gaining increasing attention in environmental, social, and cultural contexts (Grechinski, 2020). This aligns with Sustainable Development Goal 14 (SDG-14), which aims at conserving marine environments by preventing and reducing marine pollution by 2030 (United Nations, 2022).

The purpose of this study was to assess the levels of cigarette butt contamination and their distribution over time and space. Additionally, the study aimed at determining the degradation levels of the butts and audit the brands of the material found on beaches with varying levels of use. This information will serve as a basis for developing measures and actions to control and reduce the problem.

2. Materials and methods

The study was carried out in two urban beaches along the coast of Recife-PE (Northeast Brazil) with varying levels of usage: Pina beach (P1: 8°05′28.71″S / 34°52′51.90″W) and Boa Viagem beach (P2: 8°05′50.20″S / 34°52′57.15″W). Although both beaches are situated in well-preserved environments with good leisure infrastructure, they have different patterns of usage. P1 (highly used) has a frequency of up to 3,020 users/100 m of beach, while P2 (less used) has a frequency of 516 users/100 m of beach (Silva et al., 2008a). Collections were made before the arrival of vendors and bathers, always on Sundays at 5am, to ensure standardisation and because it was the day when the areas were exposed to the highest level of use (Silva et al., 2008a). The collections took place during the winter (3 in May, 2 in July, and 1 in August 2021) and summer (3 in November 2021 and 3 in January 2022).

Ten transects (1.5 m width) were marked, with intervals of 10 m between them, extending from the Solarium area (Polette and Raucci, 2003) to the highest watermark (Fig. 1). Each transect was calculated individually, as the strandline in P1 and P2 had irregular markings. The density was determined based on the total area of the transects in both regions.

Following the collections, all materials visible on the surface and up to a depth of 2 cm were gathered and sent for screening.

During the screening process, mass and length analyses, degradation level assessments, and brand classifications were conducted. To achieve this, all materials were dried in an oven at 60 °C until their dry mass stabilised, reducing water retention and minimising its influence on the mass result. All cigarette butts that still had brand markings were examined for brand audit analysis (Baxter et al., 2022), which helped identify the primary cigarette litterers at the study site. Additionally, the audit enabled the evaluation of the brands permitted for sale and consumption in the country through the annual report of the National Health Surveillance Agency (*Agência Nacional de Vigilância Sanitária* - ANVISA).

To determine the stages of degradation, there was a screening with the methodology described by Araújo et al. (2022), that identify four types of degradation of cigarette butts. Type I includes butts with preserved filter and paper coating, as well as the presence of tobacco. Type II includes filter and coating that are also preserved, but without tobacco. Type III includes worn paper coating (but still present) and no

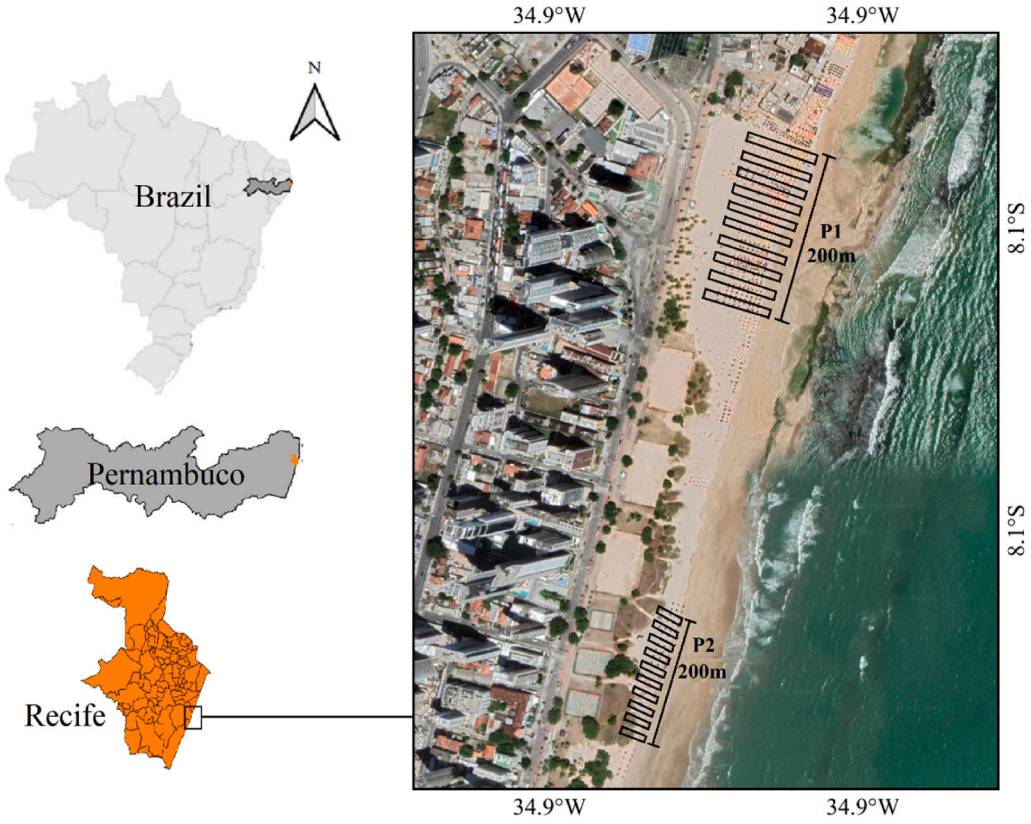


Fig. 1. Cigarette butt collection site with the delimitation of the areas according to the beach use pattern established by Silva et al. (2008), Pina (P1: much used) and Boa Viagem (P2: little used) beaches, Recife-PE. Within the delimited areas, 10 1.5 m wide transects were laid out in a south-north direction at 10 m intervals, going from the Solarium area (Polette and Raucci, 2003) to the highest watermark.

tobacco. Type IV includes only the filter, without tobacco and paper coating (Fig. 2). We handled all materials during sorting while wearing Personal Protective Equipment (PPE).

To represent the sampling areas, the number of cigarette butts was standardised per square metre. To analyse significant differences ($p < 0.05$) in butts/m^2 between areas with different occupancy configurations, collections, transects, and seasons of the year, the Kruskal-Wallis test was used. The Pearson linear correlation coefficient was used to analyse the correlation between two variables. BioEstat 5.4 was used for the analyses.

3. Results and discussion

3.1. Density of cigarette butts

A total of 10,275 cigarette butts were found in the two collection areas (Fig. 3a). As expected, the number and density of cigarette butts were directly proportional to the level of beach use. In the most heavily used beach (P1), 9,958 cigarette butts (96.91% of the total collected) were counted within an area of 1,125 m^2, resulting in a density of 8.85 cigarette butts/m^2. In P2, the beach with lower intensity of use, the number of cigarette butts totalled 317 in an area of 300 m^2, resulting in 1.05 cigarette butts/m^2. The density of cigarette butts on beaches can vary greatly around the world. To compare cigarette butt contamination on beaches in Northeast Brazil and coastal areas worldwide (Table 1), an evaluation indicator for plastic debris on beaches developed by Alkalay et al. (2007) was used as a basis. The measure chosen as numerator of the index approximates the size of cigarette butts (>2 cm), with the information of the items collected by establishing transects. However, even with an indicative of 50% for Northeast Brazilian beaches and 33% for globally classified extremely polluted beaches, there may still be underreporting due to some cigarette butts being smaller than the size used in the indicator. This study used the method for the purpose of comparing scientific studies on cigarette butts, despite its limitations, as it only uses the density of plastic debris as a variable for its definition. The choice of this method was due to the lack of standardised information related to the density of cigarette butts. Another known method is the Cigarette Butt Pollution Index (CBPI) developed by Torkashvand et al. (2021) for public areas. The CBPI calculation (CBPI = Cigarette Butt Density × E) includes a coefficient E, which is a predetermined summation of variables: (1) soil condition, (2) type of urban roads, (3) annual precipitation (mm), and (4) distance to groundwater (m). However, it was not possible to apply this method in the present study as two variables (2) and (4) do not include information from the study areas and therefore would not provide a real value for coefficient E.

A significant difference ($p < 0.05$) was observed between the average number of cigarette butts per square metre in P1 and P2 (P1 > P2). This variation may be attributed to the dynamics of the collection sites. In addition to having a smaller sand strip with less occupation influence, P2 may also have more effective cleaning services compared to P1′s sand strip.



Fig. 2. Different levels of cigarette butts degradation proposed by Araújo et al. (2022).

In certain marine regions, the contamination of cigarette butts exceeds that of urban areas, as demonstrated by Nasab et al. (2022) study that found a density of 0.25 butts/m^2 in parks and 0.31 butts/m^2 on no urban. This discrepancy can be even greater in marine zones located near central regions or areas with great influence from beachgoers. By examining the scientific studies presented in Table 1, which document the presence of cigarette butts in the Metropolitan Region of Recife, it was discovered that over 40,000 cigarette butts were improperly discarded in the coastal marine environment of Recife (Pina and Boa Viagem), representing 66% of the collected material (Table 1). These figures highlight an ongoing problem in urban beaches of Recife, underscoring the high contamination potential of cigarette butts in regions influenced by high tourism (MAIA et al., 2014). The concentration of cigarette butts in these areas, besides being influenced by human occupation and the inefficiency of cleaning services that is not efficient in removing this material (Magalhães and Araújo, 2012), can also be attributed to negligence arising from lack of awareness about its potential for contamination.

There was a significant difference ($p < 0.05$) in the number of cigarette butts per square metre between transects in P1, but not in P2 ($p > 0.05$). Critical areas were observed during collection moments in P1, particularly in transects 6 to 10, where there was a higher density of cigarette butts per square metre (Fig. 3d). These areas were characterised by the presence of bars, chairs, and informal vendors.

This distribution reveals that the disposal of cigarette butts is closely linked to human activity, with the number of users having a direct impact on the amount of waste generated. This information can be used to develop sustainable practices for marine environments, including measures to reduce critical points, especially when dealing with toxic materials. Additionally, in light of previous studies on the high density of cigarette butts (Simeonova et al., 2017; Silva-Cavalcanti et al., 2013), it is possible to classify them as flag items (Silva et al., 2008a) and recognise their importance in monitoring mesolitter.

In terms of seasonality, although there was a higher average density of cigarette butts during the summer period (Fig. 3b), there was no significant difference ($p > 0.05$) between seasons for P1. This section covers the highest occupancy rate in both seasons (SILVA et al., 2008b). However, there was a significant difference ($p < 0.05$) between seasons for P2, with a higher occurrence of cigarette butts during the summer period. These values suggest that cigarette butts on urban beaches with different levels of use follow the same distribution pattern as other litter, indicating that they are highly influenced by human occupation (Burlat and thorsteinsson, 2022; Asensio-Montesinos et al. 2021; Nachite Et Al. 2019; Simeonova et al., 2017).

In terms of collection days, a high number of cigarette butts were observed over the course of twelve Sundays (Fig. 3d), with a significant difference ($p < 0.05$) between the dates. During collections in the rainy season, the presence or absence of precipitation affected the detection of cigarette butts. Sometimes, the grains adhere to butts on the surface, blending in with the sand and making them difficult to identify. Therefore, extra care is needed when looking for cigarette butts in wet sand.

The Pearson correlation analysis revealed that precipitation had an impact on the density of cigarette butts. In P1, a moderate negative correlation was observed between collection days ($r = -0.46$), where dates with higher precipitation levels had lower values of cigarette butts. In P2, a weak negative correlation was observed ($r = -0.31$), where rainfall had less influence on the density of the material in the region. Tourist areas, such as beaches, tend to have fewer users on days with high rainfall, resulting in a decrease in the number of discarded items. This behaviour was observed during the collections, as locations with high precipitation levels on had fewer users.

The quantification of cigarette butts can aid in determining contamination levels in marine environments. However, this requires standardisation of studies that monitor litter in these regions. Factors such as the number of butts, total area analysed, and days of collection

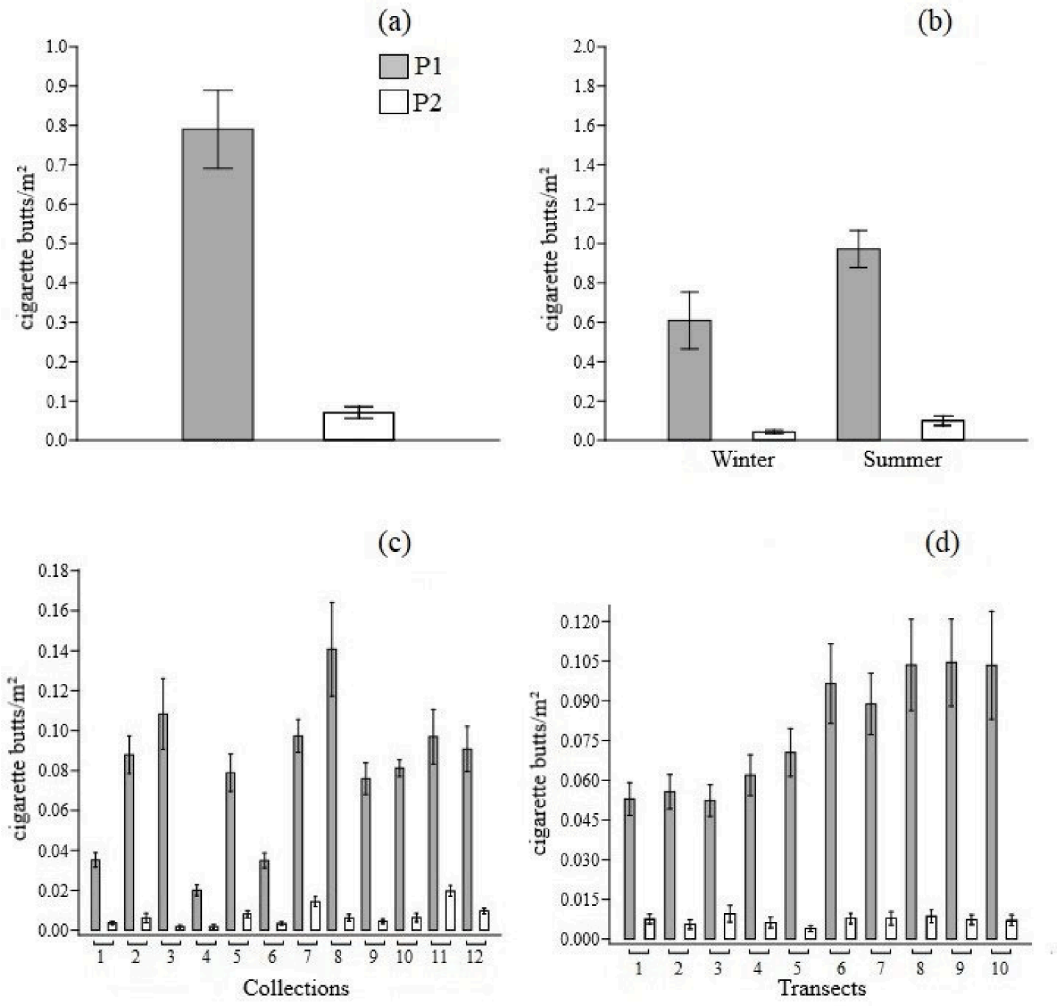


Fig. 3. Mean and standard deviation of cigarette butts/m^2 collected between May 2021 and January 2022 at the evaluated beaches. (a) Average and standard deviation of cigarette butts/m^2 in areas with different use patterns. (b) Average and standard deviation of cigarette butts/m^2 according to seasonality. (c) Average and standard deviation of cigarette butts/m^2 along the collection days and (d) Average and standard deviation cigarette butts/m^2 according to transects.

can affect the determination of beach contamination levels. Nevertheless, there is a lack of consistency in the dissemination of these results among scientific studies.

3.2. Degradation of cigarette butts

All four stages of degradation were observed during the 12 collections. A significant difference ($p < 0.05$) was found between the types of cigarette butts in the two study areas, with type IV being the most common in both P1 and P2 (Table 2).

Type I cigarette butts contain tobacco that can easily spread and contaminate the environment (Araújo et al., 2022). Additionally, their yellowish appearance is due to the filtration of compounds during smoking, while type IV butts have a whitish appearance due to environmental processes. Farzadkia et al. (2022) report that 147.5 kg/year of heavy metals are released from cigarette butts into the environment, potentially causing toxic effects on local biodiversity through leaching. Apart from the degradation levels, it was observed that type IV butts showed significant flattening of fibres, with mean and standard deviation (mm) varying between: type I: 8.25 (±0.76), type II: 8.05 (±0.84), type III: 7.67 (±0.92), and type IV: 6.41 (±1.07). This flattening process may make penetration into the soil and camouflage more efficient, potentially hindering their removal from the soil.

In terms of morphometry, a significant difference ($p < 0.05$) was observed in the length and mass of cigarette butts. The highest values were found in freshly discarded or Type I butts (Fig. 4). Although some items exceeded 2.5 cm in size, qualifying as macro litter, most of their length is below this value. The length of this material has average values with variations between types (Fig. 4), with Type I measuring 2.70 cm (±0.32), Type II measuring 2.22 cm (±0.31), Type III measuring 2.19 cm (±0.23), and Type IV measuring 2.21 cm (±0.20), with their dimensions predominantly restricted to mesolitter (5 mm to 2.5 cm).

The difference between types I and IV may be attributed to the process of fibre detachment, which is likely caused by friction with beach sand (Araújo et al., 2022). Puls et al. (2011) suggest that the complete degradation of conventional cigarettes is hindered by the high compaction of fibres and the addition of plasticisers. Furthermore, the duration of exposure is a crucial factor, as longer exposure times lead to more severe damage.

Considering that cigarette butts are non-biodegradable and highly resistant to physical degradation processes, the magnitude of their impact due to the presence of hundreds of toxic compounds is evident (Araújo et al., 2022). The loss of the paper coating leads to a clear modification in the filter, with evidence of fibre loss and gradual reduction in volume. In this study, a reduction of approximately 40% in total mass was observed when comparing cigarette butts of type I and IV. Bonanomi et al. (2015) found a reduction of 37.8% in comparison to the initial mass over two years, suggesting that the 5,132 type IV cigarette

Table 1
Values of litter on world and Brazilian beaches, with indication of collection sites, density of litter and proposed visual assessment for plastic particles larger 2 cm in size prepared by Alkalay et al. (2007)*.

REFERENCE	LOCATION	CIGARETTE BUTTS/M²	VISUAL SURVEY
	WORLD		
ASENSIO-MONTESINOS, et al. (2021)	**SPAIN**		
	Cádiz	0.040	Very clean
	Ceuta	0.071	Very clean
	Alacant	0.084	Very clean
NASAB et al. (2022)	**IRAN**		
	Beach 1	0.05	Very clean
	Beach 2	0.17	Clean
	Beach 3	0.09	Very clean
KATARŽYTĖ et al. (2020)	**GERMANY****	0.86	Dirty
	LITHUANIA**	0.013	Very clean
NACHITE et al. (2019)	**MOROCCO****	0.434	Moderate
LOIZIDOU, LOIZIDES & ORTHODOXOU, (2018)	**CYPRUS****	0.21	Clean
KUNGSKULNITI et al. (2018)	**THAILAND**		
	Bang Saen	0.62	Dirty
	Cha am	0.62	Dirty
	Mae Rum Pueng	0.98	Dirty
	Laem Mae Pim	2.30	Extremely dirty
	Jao Lao	0.25	Moderate
	Banchuen	13.30	Extremely dirty
	Sairee	1.05	Extremely dirty
	Bohpud	0.99	Dirty
	Chaloke-Ban Kao	0.45	Moderate
	Chalatas	3.56	Extremely dirty
	Patong	0.76	Dirty
LAGLBAUER et al. (2014)	**SLOVENIA**		
	Debeli Rtič	3.24	Extremely dirty
	Jadranska-ska	2.88	Extremely dirty
	Simonov Zaliv	20.24	Extremely dirty



(*continued on next page*)

Table 1 (*continued*)

				30
	Bele Skale	0.16	Extremely dirty	
	Portorož	6.94	Extremely dirty	
	Seča	0.3	Moderate	
MUNARI et al. (2016)	**ITALY**			
	Rosolina	0.026	Very clean	
	Bellocchio	0.020	Very clean	
	Casalborsetti	0.029	Very clean	
	Bevano	0.043	Very clean	
	Volano	0.14	Clean	
	BRAZIL			
	PERNAMBUCO			
Current study	Pina	8,85	Extremely dirty	
Current study	Boa Viagem	1,05	Extremely dirty	
DIAS-FILHO et al. (2011)	Boa Viagem	2,84	Extremely dirty	
SILVA-CAVALCANTI, ARAUJO & COSTA (2013)	Boa Viagem	3,10	Extremely dirty	
ARAÚJO & COSTA (2021)	Maria Farinha	0,67	Dirty	
	Bairro Novo	0,93	Dirty	
	Piedade	0,36	Moderate	
	Porto de Galinhas	1,56	Extremely dirty	
	RIO GRANDE DO NORTE	1,41	Extremely dirty	
	Jenipabu			
	Artistas	0,93	Dirty	
	Pirangi	0,63	Dirty	
	Ponta Negra	0,42	Moderate	

(*) Reference values based on visual assessment of plastic debris in coastal areas by Alkalay et al. (2007), which states: 0–0.1 items/m^2 – very clean; 0.1–0.25 items/m^2 – clean; 0.25–0.5 items/m^2 – moderate; 0.5–1 items/m^2 – dirty; more than 1 items/m^2 – extremely dirty.
(**) The cigarette butts/m^2 values from the sum of the study areas and not with information of cigarette butts/m^2 per beach, since this information is not in the reference.

Table 2
Mean and standard deviation of cigarette butts/m^2 and their degradation levels. Demonstration of degradation levels from the types (I, II, II and IV) established by Araújo et al. (2022). Values of cigarette butts/m^2 of the items collected on the beaches of Pina and Boa Viagem (Recife-PE) in relation to their stages of degradation.

TYPES	P1	P2	%
I	0.076 (±0.078)	0.015 (±0.013)	10.49
II	0.146 (±0.113)	0.016 (±0.021)	18.71
III	0.165 (±0.087)	0.011 (±0.012)	20.86
IV	0.397 (±0.177)	0.027 (±0.016)	49.94

butts found in the study areas are contaminating the environment for about two years. Further studies are necessary to estimate the long-term degradation of cigarette butts, including the time of their disappearance in the environment and the actual effects of their fibres.

3.3. Brand audit

A total of 18 cigarette brands were identified in 3,434 cigarette butts (33.42%). All brands were found in P1, while only 50% were found in P2 (Table 3), indicating a significant difference ($p < 0.05$) in brand distribution between the study locations. The cigarette butts also varied in type, including traditional, mentholated, and ventilated filter butts. Depending on the type, cigarettes can contain different amounts of compounds, with menthol cigarettes having a higher nicotine content than traditional ones (Parker and Rayburn, 2017), which may increase environmental contamination.

The most predominant cigarette brand in both areas was brand **(A)**, with 1.40 cigarette butts/m^2 (1,483 items) in P1 and 0.13 cigarette butts/m^2 (58 items) in P2. According to the ANVISA annual report (2022), there are a total of 11 legal cigarette brands, 4 illegal brands, and 3 that have undergone name changes or are no longer manufactured but still have their circulation permitted (Table 3). Lima et al. (2021) observed the occurrence of 22 cigarette brands in 53.94% of the cigarette butts collected in Santos-SP. Companies that produce tobacco-derived products are taking sustainable measures, such as educational activities, distribution of ashtrays, and partnerships with cleaning campaigns. However, this effort is interpreted as a strategy to transfer responsibility to consumers (Lima et al., 2021). One alternative for reducing impacts is through environmental management and reverse logistics, which involves product management by manufacturers even after the material is used (Santos, 2012).

According to ANVISA (2022), in Brazil, eight companies are registered for smoking products. Out of these companies, four are responsible for producing the items found on the evaluated beaches. The first produces five brands, the second produces two brands, the third produces two brands, and the fourth also produces two brands. It was observed that the companies follow a pattern when disclosing ingredient information, using the same text for all types of cigarettes/brands produced by the same company. Therefore, if cigarettes A, B, C, and D are of different types, they will present the same ingredient information since they are all produced by company 1 (Table 3). As a result, many pieces of information, especially regarding compound concentrations, are concealed and made invisible due to the small font size used.

Despite pressure from the tobacco industry, some countries have implemented restrictions on smoking in enclosed spaces (Daube and Maddox, 2021). In New Zealand, stricter measures are being implemented to reduce the number of smokers, with the goal of having less than 5% of the population smoking by 2025, and future generations will no longer be able to purchase cigarettes (Agrawal and Britton, 2022). These actions have the potential to significantly reduce the negative impact of smoking on public health and the environment, and can serve as a model to be followed by other countries, with one example being Mexico where smoking is no longer allowed in some public places (EL PAÍS, 2023).

According to information on retail prices provided by the Brazilian Federal Revenue in 2022, the manufacturer of cigarette **(A)** sells a pack of their product for R$ 6.25 (US$ 1.27) in Pernambuco, while the second most commonly found brand, cigarette **(B)**, is sold for R$ 9.00 (US$ 1.83) in the same state. Users in this region have a lower purchasing power (Araújo et al., 2012), that may lead them to choose cheaper brand products. The low cost of the product may make it easier to purchase, consume, and dispose of. In 2020, it was reported that 49% of cigarettes consumed in Brazil were illegal, which are cheaper than formal market cigarettes (ETCO, 2022).

The ANVISA annual report (2022) on sales and consumption in the country reveals that Pina and Boa Viagem beaches account for 61% of brands with authorised sales, 22% of illegal brands, and 17% of brands that have either changed names or are no longer in production. To reduce the consumption of illegal cigarettes and minimise their environmental impact, strategies should be implemented to eliminate their sale and decrease demand (Goodchild et al., 2022).

Information related to control, regulation, and proper disposal can aid in reducing the issue of contamination in urban beaches. Despite scientific studies on cigarette butts, which indicate that 29.2% of publications focus on damage to biodiversity, 25.5% on public policies, 24.3% on contamination of butts in public environments, and 21% on

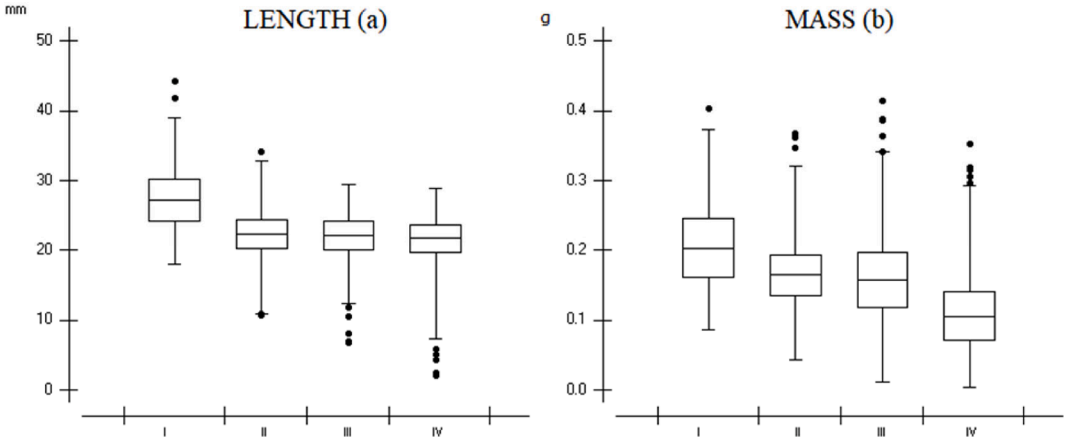


Fig. 4. Means and standard deviation of morphometry according to the degradation stage of cigarette butts at different degradation levels (type I, II, III and IV) collected from May 2021 to January 2022 at urban beaches in Northeast Brazil. (a) Mean and standard deviation of the length of the cigarette butts at different levels of degradation, with unit in millimeters. (b) Mean and standard deviation of the mass of the cigarette butts at different levels of degradation, with unit in grams. The values above and below (black points) are the outliers or discrepant points.

Table 3
Mean and standard deviation values of the cigarette brands found on the beaches of Pina and Boa Viagem, with their characteristics and compounds present on the product packaging.

BRAND	P1	P2	Characteristics	Compounds
(A)	0.25 (±0.06)	0.025 (±0.0036)	Cigarette white in color, filter ranging from white to yellow with brown streaks and red seal.	Mixture of tobaccos, sugars, adhesives, binding agents, combustion agents, flavoring agents, cigarette paper, dyes, and humectants.
(B)	0.17 (±0.04)	0.011 (±0.0016)	Cigarette white in color, filter ranging from white to brown with small white spheres.	Mixture of tobaccos, sugars, adhesives, binding agents, combustion agents, flavoring agents, cigarette paper, dyes, and humectants.
(C)	0.06 (±0.01)	0.005 (±0.0008)	White colored cigarette, white filter with two gray lines on top of the filter.	Mixture of tobaccos, sugars, cigarette paper, plant extracts, flavoring agents.
(D)	0.03 (±0.01)	0.002 (±0.0004)	White colored cigarette, white filter with several gray diagonal stripes along the filter.	Mixture of tobaccos, sugars, adhesives, binding agents, combustion agents, flavoring agents, cigarette paper, dyes, and humectants.
(E)	0.013 (±0.005)	0.003 (±0.0005)	White colored cigarette, white filter with several gray sinuous stripes along the filter.	Mixture of tobaccos, sugars, adhesives, binding agents, combustion agents, flavoring agents, cigarette paper, dyes, and humectants.
(F)	0.008 (±0.002)	0.002 (±0.00038)	White colored cigarette, white filter with a thick blue stripe on top of the filter.	Mixture of tobaccos, sugars, cigarette paper, plant extracts, flavoring agents.
(G)	0.005 (±0.001)	0.002 (±0.0004)	Cigarette white color, brown filter with small white spots.	Mixture of tobaccos, sugars, cigarette paper, plant extracts, flavoring agents.
(H)	0.004 (±0.002)	0.0004 (±0.00001)	White colored cigarette, with filters ranging from white to brown.	Mixture of tobaccos, adhesives, binders, combustion agents, cigarette paper, inks, and humectants.
(I)	0.0008 (±0.0001)	0.0009 (±0.00001)	White colored cigarette, with white filters, featuring a brown tobacco branch design.	Smokes, sugars and cigarette paper. Presenting: 7 mg of tar, 0.6 mg of nicotine and 8 mg of carbon monoxide.
(J)	0.0017 (±0.001)	–	White colored cigarette, with filters varying between white and brown, with a drawing of an animal.	Mixture of tobaccos, sugars, adhesives, binding agents, combustion agents, cigarette paper, flavorings, preservatives, inks, and humectants.
(K)	0.00020 (±0.0004)	–	White colored cigarette, with white filter, showing two thin blue stripes on top of the filter.	Tar: 8 mg Nicotine: 0.5 mg Carbon monoxide: 10 mg
(L)	0.00017 (±0.0003)	–	White colored cigarette, with white filter, showing two blue stripes, one thin and one thick on top of the filter.	Mixture of tobaccos, sugars, cigarette paper, plant extracts, flavoring agents.
(M)	0.0003 (±0.0001)	–	Cigarette white in color, with filters ranging from brown to white.	Mixture of tobaccos, sugars, adhesives, binding agents, combustion agents, flavoring agents, cigarette paper, dyes, and humectants.
(N)	0.0012 (±0.001)	–	Black color cigarette, black filter, with silver stripe on top of the filter.	Cigarette without composition information.
(O)	0.0002 (±0.00001)	–	Cigarette white in color, with filters ranging from white to red.	Cigarette without composition information.
(P)	0.0003 (±0.00001)	–	White colored cigarette, with white filter.	Mixture of tobaccos, sugars, cigarette paper, plant extracts, flavoring agents.
(Q)	0.0002 (±0.00001)	–	Cigarette white color, brown filter with small white spots.	Tobacco blends, cigarette paper, combustion agent, adhesive, preservatives and humectants.
(R)	0.0003 (±0.00001)	–	White colored cigarette, with white filter, with silver stripe on top of the filter.	Cigarette without composition information.

the physical and chemical characteristics of the material (Silva et al., 2023), there is still a lack of practical solutions and actions to reduce contamination in these environments.

Therefore, manufacturers who produce products but neglect their proper disposal must implement reverse logistics. So far, the municipal government in the studied areas has not taken any measures to reduce or eliminate the presence of cigarette butts on urban beaches.

3.4. Management of cigarette butts

The distribution of people on beaches is strongly linked to the high number of material discarded in areas (Magalhães and Araújo, 2012), also occurring with cigarette butts, and the data from the present work can serve as a basis for improving the problem on the beaches of Recife or adjacent regions. It is important to have a more direct approach with beach goers, about the impacts of incorrectly discarded cigarette butts, since these items are not removed through regular beach cleaning, and it is necessary that users be constantly warned about this issue. The implementation of actions with the objective of environmental

education for users and merchants with a focus on toxic materials, such as the cigarette butts, is also important, since this item is usually invisible to the public authorities and users. In addition, measures with low installation and maintenance costs, such as the implementation of appropriate places for the disposal of cigarette butts at different points on the beaches (especially at critical points) are also indicated to help reduce the accumulation of the material in the region.

Although waste is composed of various materials and each one generates a consequence for the environment, it is necessary to minimize the impacts of the most harmful ones. The creation of projects focused on cigarette butts can help reduce not only this item, but also another aggravating factor present in several regions, which would be the fragmented materials that have similar sizes to those of the cigarette but are hardly collected during cleaning. The creation of a standardization scale for monitoring through data generation could also reduce the underreporting of similar sized cigarette butts and materials.

The link between the municipal government together with environmental groups and anti-smoking projects can be the key to improving human and environmental health in regions with large concentrations of cigarette butts and smokers. This discourse needs to be linked to the implementation and enforcement of laws by the powers that be that can act locally in order to reduce the problem. An example is Spain, which through the creation of a law passes the cost of cleaning up litter discarded in public places to the companies that produce the product (BBC, 2023); Mexico, where smoking is no longer allowed in some public places (EL PAÍS, 2023); and New Zealand where a minimum smoking age law was passed with the goal of banning smoking in the next generation (The guardian, 2023). In Brazil, the incorrect disposal of this material could be reduced if the Law 12.305 of 2010 of the National Solid Waste Plan (PNRS) was complied with, which determines the prohibition of disposal of solid waste on beaches, sea, and water bodies. Furthermore, the polluter principle established by the Organization for Development and Economic Cooperation (ODEC) reports that the polluter should bear the cost of pollution (Oliveira et al., 2019) with companies (or prevailing polluters) being able to defray the cost of damage and removal of material from the environment.

4. Conclusion

The disposal of waste in marine environments varies depending on the location, occupation of the area, collection methods, and season. However, certain factors such as inefficient cleaning services, large waste size, human activities, lack of environmental education, inadequate disposal sites, and manufacturers' lack of commitment exacerbate the problem of highly toxic waste that tends to aggregate.

Regarding the quantification of litter in areas with different configurations of use, it was possible to identify that several transects, mostly in the P1 area, were classified as critical points, which may require more attention during the cleaning periods or even to intensify the collection points with more functional garbage cans in these locations. Brazil has suffered from this environmental problem for more than ten years. When comparing the current study with previous surveys conducted at Boa Viagem and Pina beach, there was no improvement in the treatment and disposal of the material, and it is possible to observe an increase of more than 60% of the material found at Pina and Boa Viagem beaches over the years.

Another factor in relation to degradation indicates its residence time of two years or more, since the types of cigarette butts with the highest index were those of type IV. This can also indicate a possible physical and chemical pollution at the site, since the cigarette butts in advanced stages of degradation usually have less mass, due to the detachment of the fibers, and less contaminants through release through the leachate.

The **A** and **B** brands were indicated as prevalent pollutants on the Pina and Boa Viagem beaches, indicating about 14% of all material found. A control is necessary from the production to the destination of the post-consumption material with more effective public policy measures that could promote a reduction of damage caused by waste. The polluter pays principle could also be applied, correctly transferring the responsibility for its impacts to the companies.

It is essential that local governments recognize the extreme risks of cigarettes and their by-products, beyond the health problems, with the development of more sustainable actions, since the cigarette butts can be considered a relevant emerging pollutant with short and long term effects, still little known by the scientific community.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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